

November 27, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics
Engineering

Lesson title:

**Behavior and intelligence (collective): from distributed
intelligence in nature to bio-inspired algorithms and swarms
robotics**

Prof. Donato Romano (donato.romano@santannapisa.it)



Scuola Superiore
Sant'Anna

Let's consider a superorganism
and SWARM INTELLIGENCE



A honeybee colony contains up to 80,000 individuals

↓ * 960,000

76,800,000,000 neurons

Ant colonies range in size from a few tens of individuals to over 1 million individuals

↓ * 200,000

200,000,000,000 neurons

Desert locust swarms (gregariousness) of 80 million individuals

↓ * hundreds of thousands

>10¹³ neurons

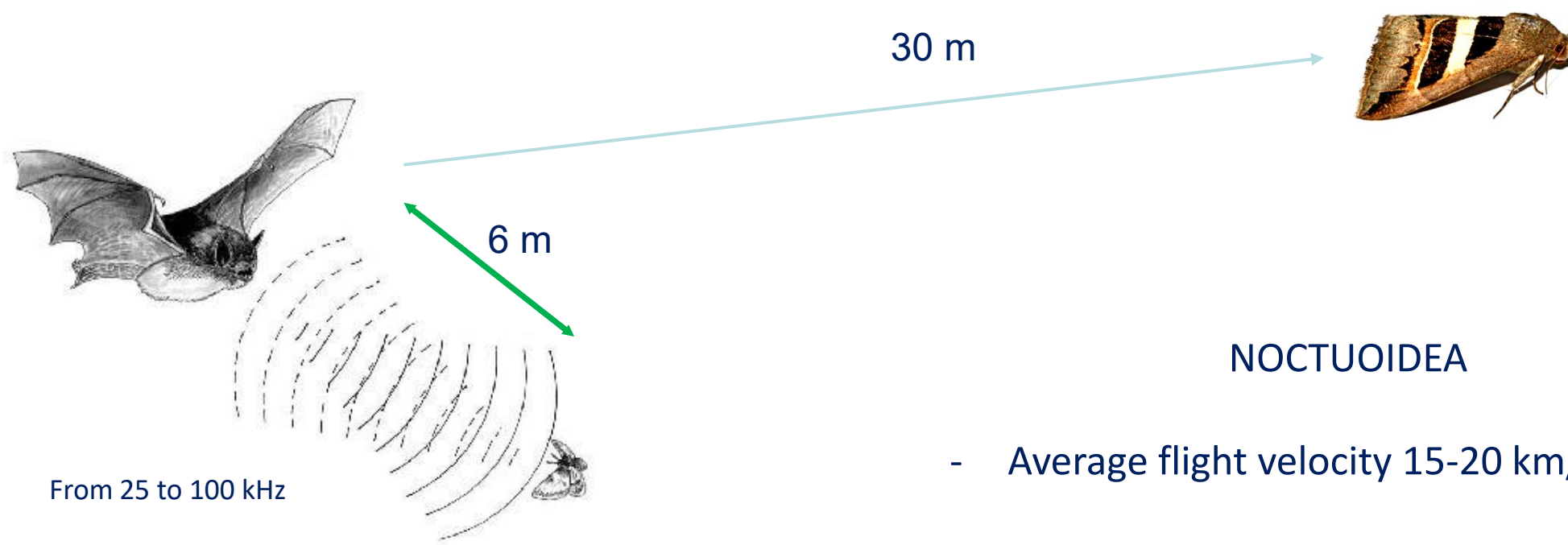
Helping Behaviour in *Megaponera*

The first recorded non-human species to consistently take care of the wounds of other individuals

- Cooperative self-defence (outside of the nest)
- Saving injured individuals
- Treatment of injured ants



Interspecific countermeasure strategies



- Average flight velocity 15-20 km/h.
- Tympanic organ located on the thorax.
- Escape velocity > 50 km/h.
- Fall to the ground.



Interspecific countermeasure strategies

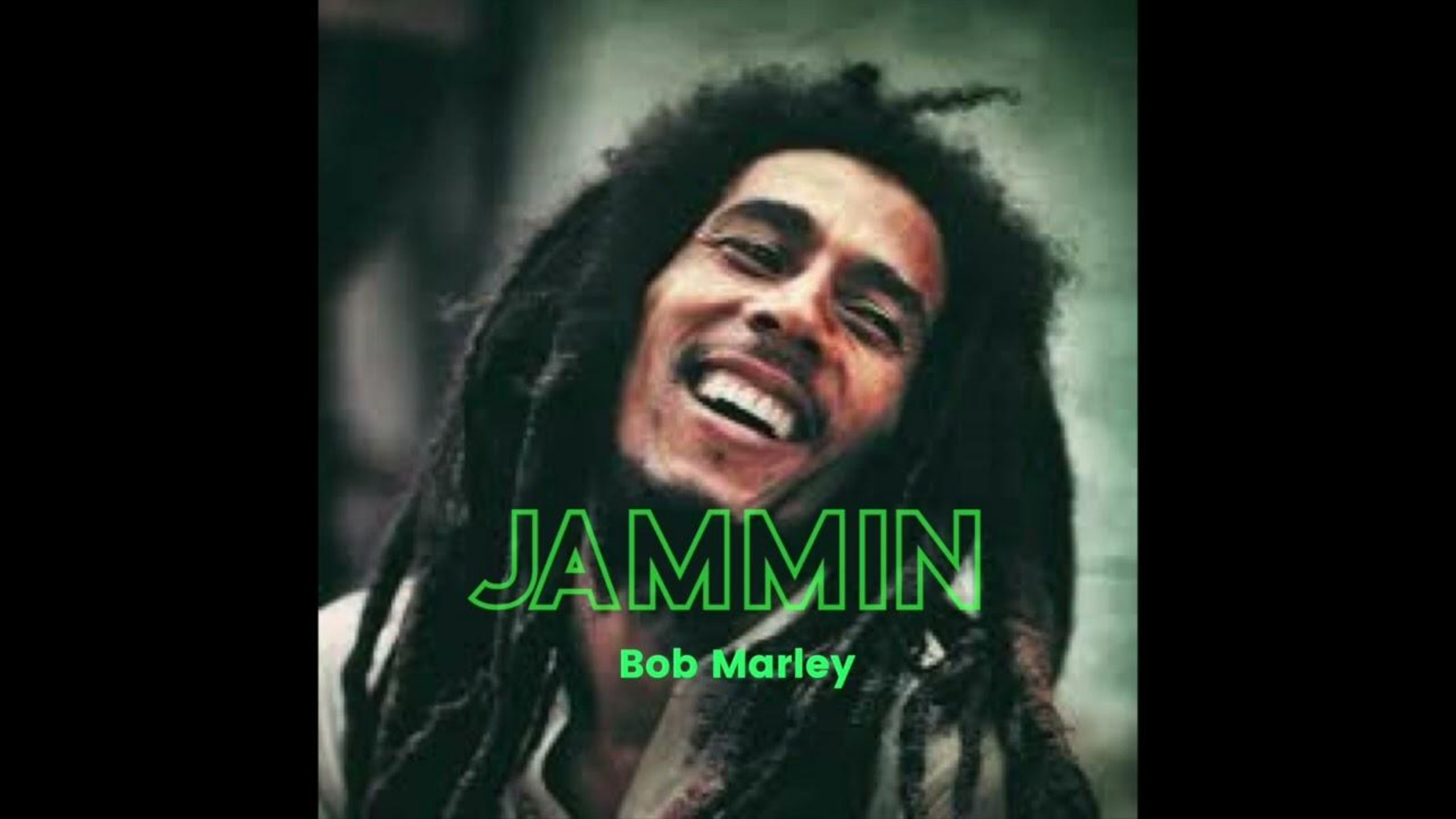


No. 360 Squadron RAF

Melese laodamia Druce (Lepidoptera: Erebidæ)



Scale in mm

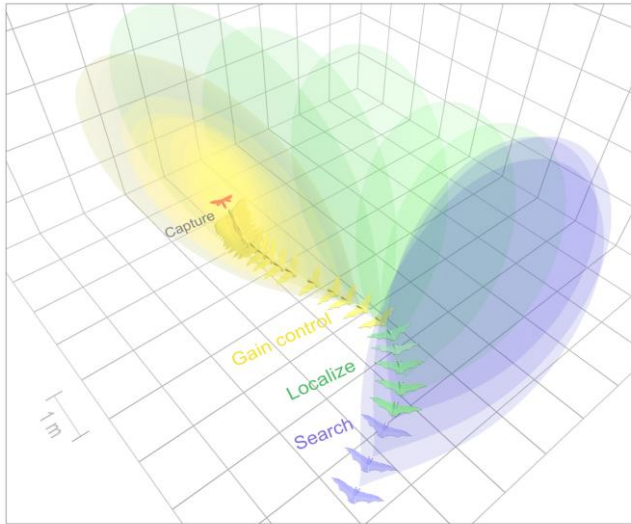
A close-up portrait of Bob Marley, smiling broadly and showing his teeth. He has long, dark dreadlocks that frame his face. The background is a soft, out-of-focus green. The word "JAMMIN" is overlaid in large, green, outlined capital letters across the lower half of his face.

JAMMIN

Bob Marley

Interspecific countermeasure strategies

- releasing a series of ultrasonic clicks that mix with the echoing signals of the bat.
- The moths generate 4,500 ultrasonic clicks per second.



Melese laodamia



Bertholdia trigona



Why is it good to live in a group?

- Mate encounters
- Anti-predator strategies
- Boosted locomotion fluid dynamics
- Searching for food
- Cooperative behaviours



What can robotics learn from animal behaviour and cognition?

- Interaction with the environment
- Learning
- Multiagent interaction
- Collective coordination
- Orientation
- ...



What is a Swarm?

A loosely structured collection of interacting agents:

Agents:

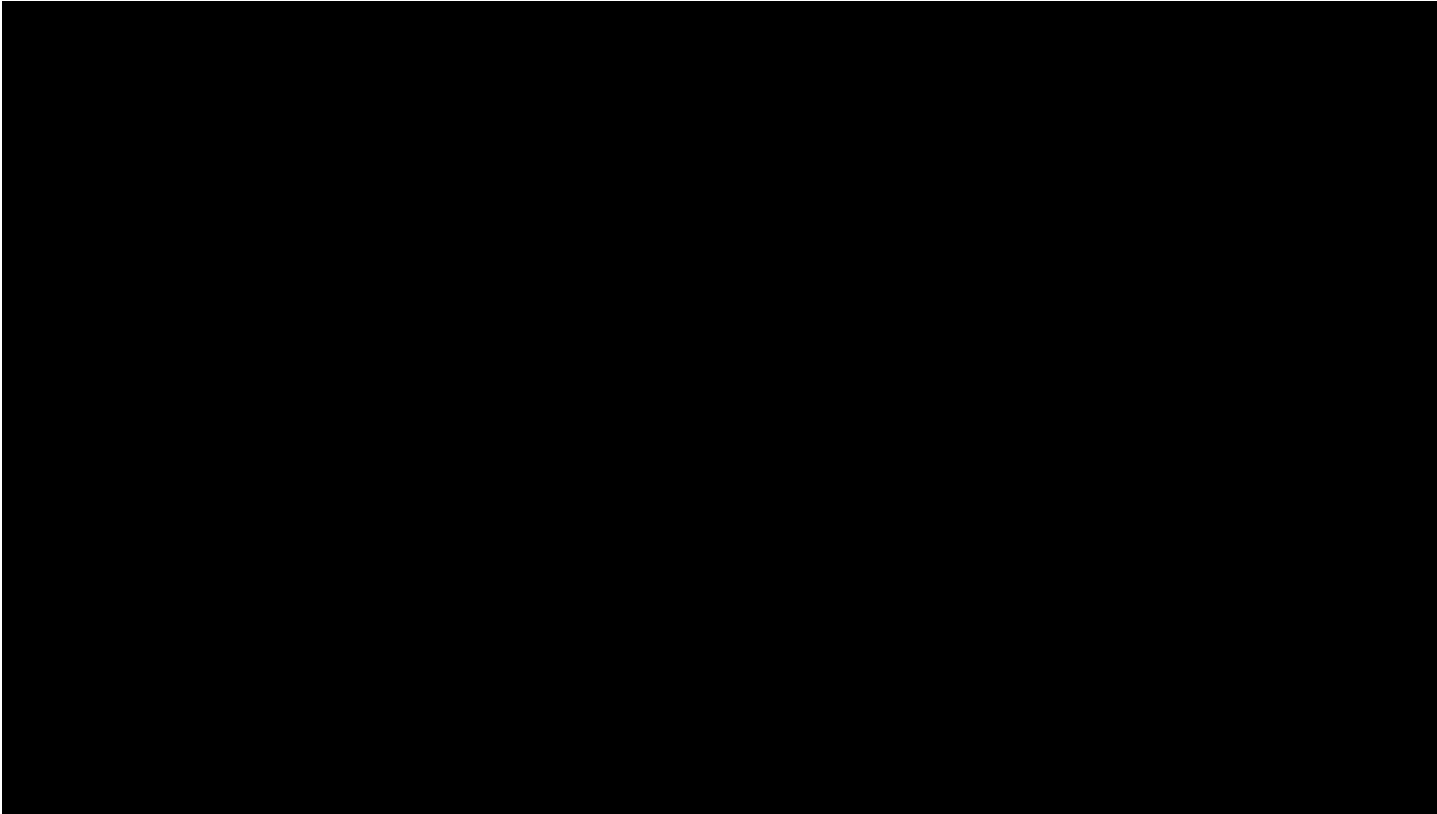
- Individuals that belong to a group
- They contribute to and benefit from the group
- They can recognize, communicate and/or interact with each other

A swarm is better understood if thought of as agents exhibiting collective behaviour



Sources of Inspiration

Bird Flocking



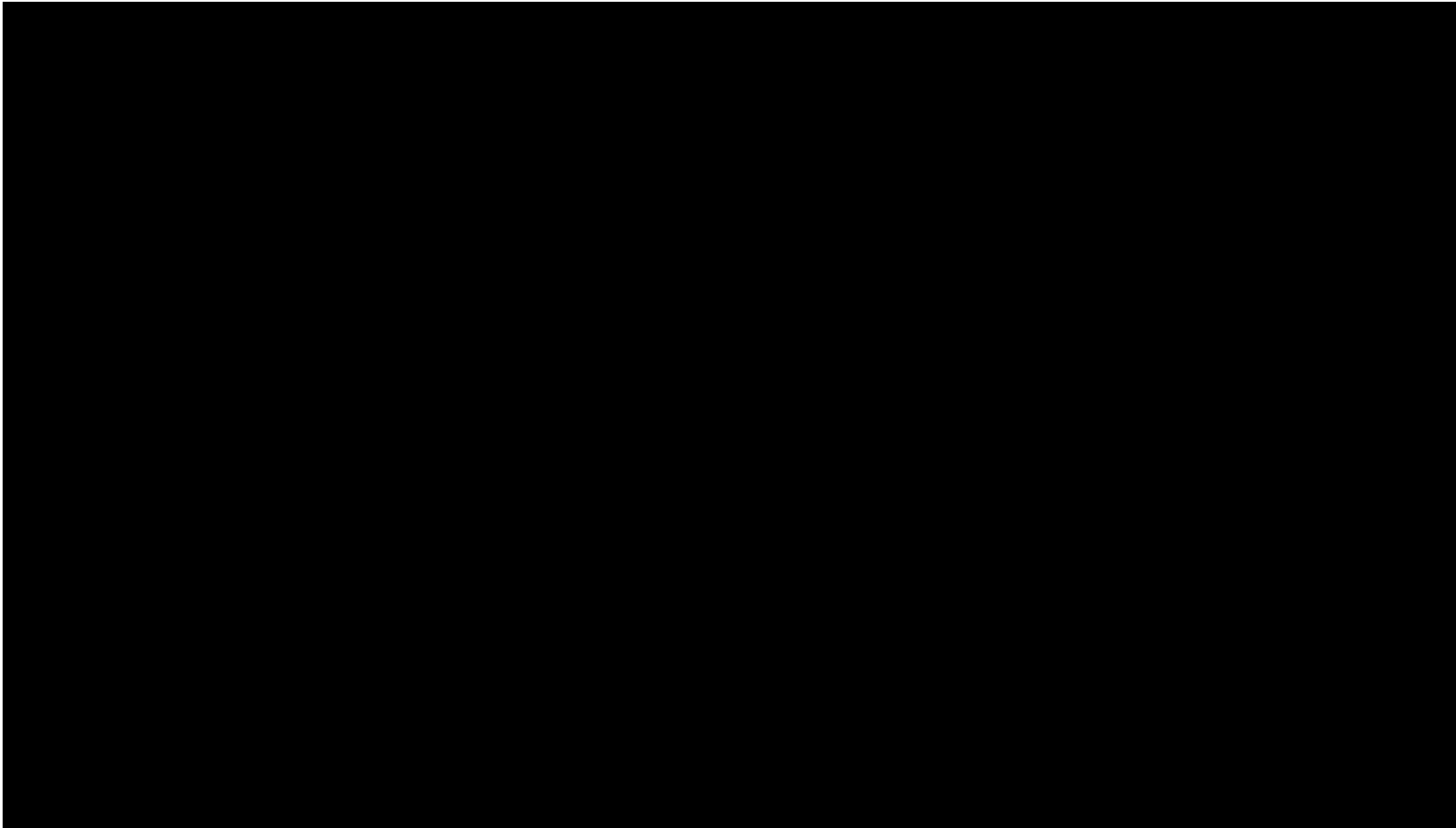
Three simple rules:

- Separation
- Alignment
- Cohesion

Birds are not following leaders during they performance, but they act as a group in which they constantly change their position



Ant colonies



Ants exhibit complex social behaviors that require group cooperation to achieve success



Fish shoal



The message passes quickly between a fish to another. The swarm follows a few simple rules: stick together, avoid collisions, swim in the same direction (schooling).

In biology, any group of fish that stay together for social reasons are **shoaling**, and if the group is swimming in the same direction in a coordinated manner, they are **schooling**. In common usage, the terms are sometimes used rather loosely. About one quarter of fish species shoal all their lives, and about one half shoal for part of their lives.

Advantages of shoaling:

- Defence against predator
- Foreging
- Finding a mate
- Hydrodynamic efficiency



Fish Schools for Smarter Wind Farms



Fish swimming in a school actually throw vortices off of their bodies as they're going. They make these spirals in the water because of their sinuous movements, but interestingly, fish behind them come up, wrap their bodies around that vortice and surf each other's wake."



This basic idea helps when examining how wind farms might be arranged to **maximize efficiency** in a similar fashion. But thanks to smart connection and interconnectivity between GE turbines, wind farms today also emulate the adaptability of fish schools.



Swarm principles

- Collective behaviour
- Local communication (direct and indirect: stigmergy)
- Self-Organization
- Division of labor
- Cooperative Transportation



Swarm principles

Collective behavior

Collective behavior refers to social processes and events which do not reflect existing **social structure**, but which **emerge** in a "spontaneous" way.

- **Complexity** from simple rules
- **Bottom up** modeling
- Individual appears as its own
- **No supervisor**
- System organized as a **whole**
- **Collective** decision making



Swarm principles

Local communication

- Self-organization in social insects often requires interactions between insects
- Such interactions can be **direct** or **indirect**
- **Direct interactions** are the "obvious" interactions: antennation, trophallaxis (food or liquid exchange), mandibular contact, visual contact, etc.
- **Stigmergy** is a method of indirect communication in a self-organizing emergent system where its individual parts communicate with one another by modifying their local environment



Stigmergy

“The coordination of tasks and the regulation of constructions does not depend directly on the workers, but on the constructions themselves. ***The worker does not direct his work, but is guided by it.*** It is to this special form of stimulation that we give the name **STIGMERGY** (*stigma*, sting; *ergon*, work, product of labour = stimulating product of labour).”

Grassé Pier-Paul, 1959

“**Indirect communication** mediated by modifications of environmental states which are **only locally accessible** by the communicating agents. ”

Dorigo & Di Caro, 1999



Stigmergy

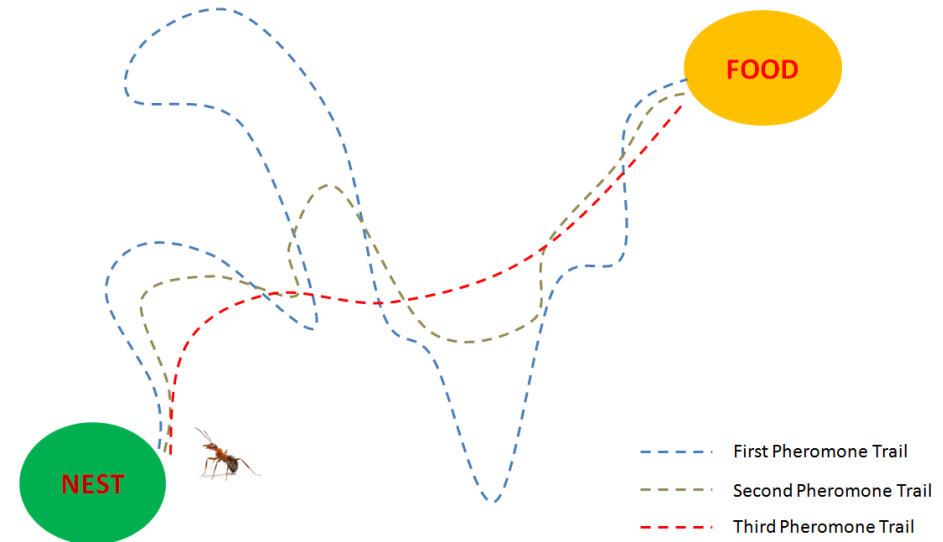
Stigmergy is a mechanism of indirect [coordination](#), through the environment, between agents or actions. The principle is that the trace left in the [environment](#) by an action stimulates the performance of a next action, by the same or a different agent. In that way, subsequent actions tend to reinforce and build on each other, leading to the spontaneous emergence of coherent, apparently systematic activity.

Stigmergy is a form of [self-organization](#). It produces complex, seemingly intelligent structures, without need for any planning, control, or even direct communication between the agents. As such it supports efficient collaboration between extremely simple agents, who lack any memory, intelligence or even individual awareness of each other.



Optimization of the colony in ants

Working as a Research and Development engineer in the 1990s telecommunications industry, Eric Bonabeau was faced with a problem. How do you successfully and efficiently route millions of communications signals across a limited amount of potential pathways?



Swarm intelligence can be applied to technology: Bonabeau and research partner Guy Théraulez noticed that ants had found a way to collectively find the quickest paths toward food using a **system of pheromones**.

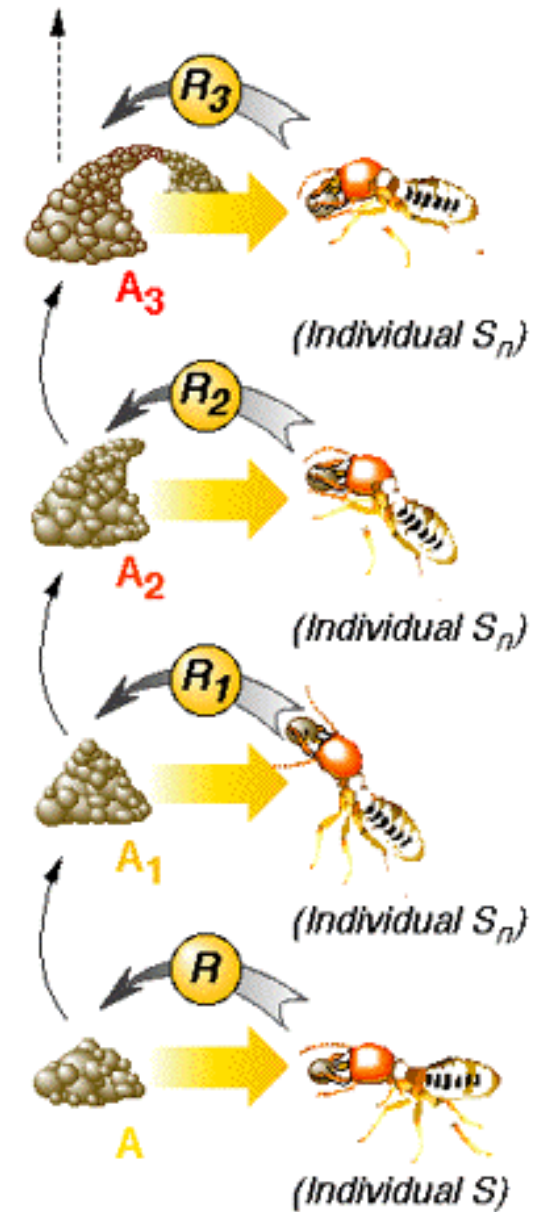
Bonabeau then took this basic idea and applied it to **communication networks**. The secret lay in creating a software “ant equivalent” that could route communications through the least congested pathways.



Stigmergy in Termites Nest

Assume that the architecture reaches state A, which triggers response R from worker S.

- A is modified by the action of S (e.g. S may drop a soil pellet), and transformed into a new stimulating configuration A1
- A1 may in turn trigger a new response R1 from S or any other worker S_n and so forth.
- The successive responses R1, R2, R_n may be produced by any worker carrying a soil pellet.
- Each worker creates new stimuli in response to existing stimulating configurations. These new stimuli then act on the same termite or anyother worker in the colony.

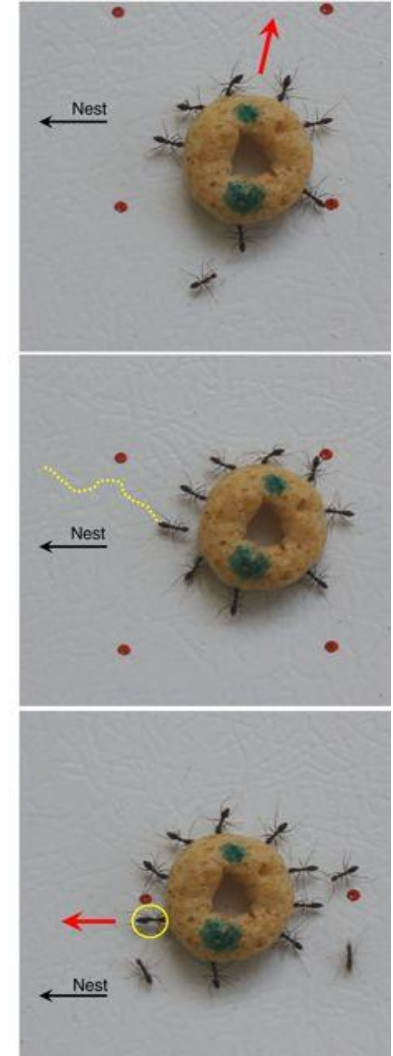
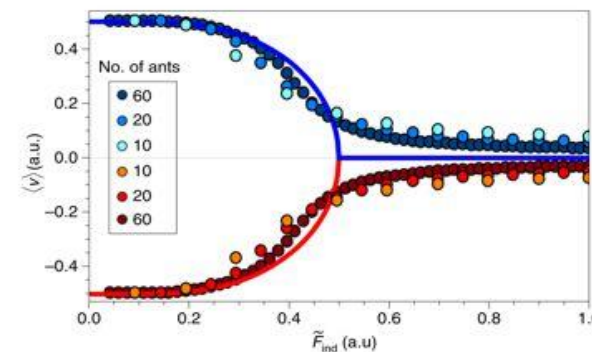
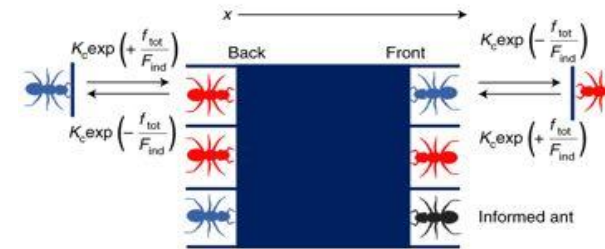
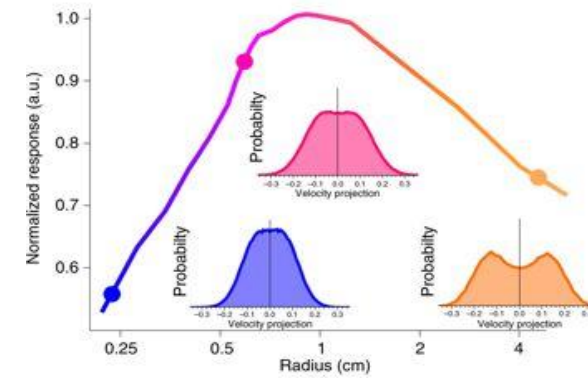


Self-organization of the colony in ants

Collective decision making



Cooperative transport



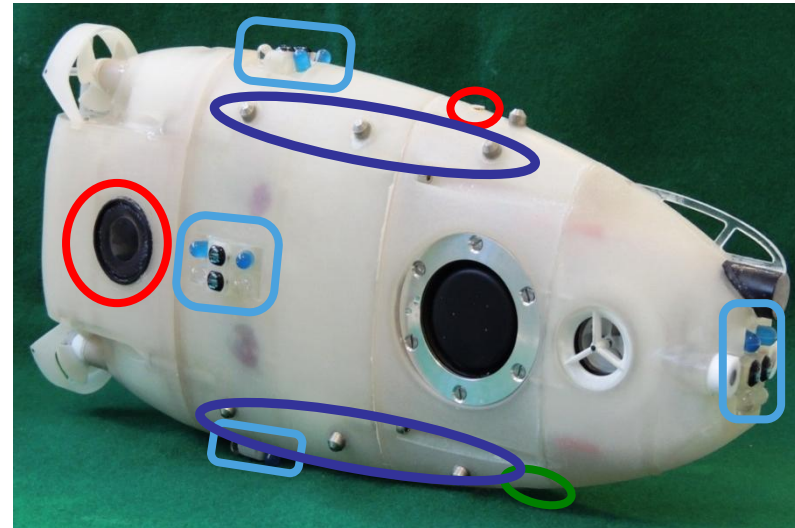


Jeff Requirements:

- **Swarm capabilities:** 3D Swimming and multiple sensing and communication
- **Cognition capabilities:** wide range of proprio and exteroceptive sensors and local communication
- **Affordable experimental costs:** to perform experiments with large swarm of AUV in a conventional swimming pool.

Main features and components

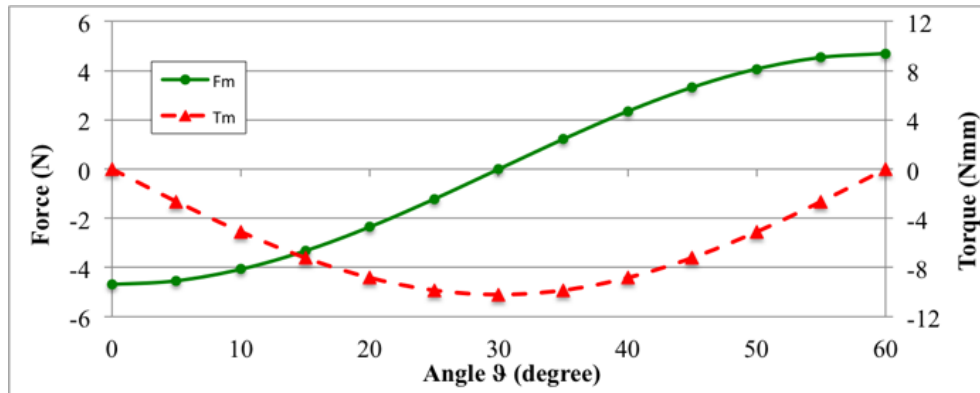
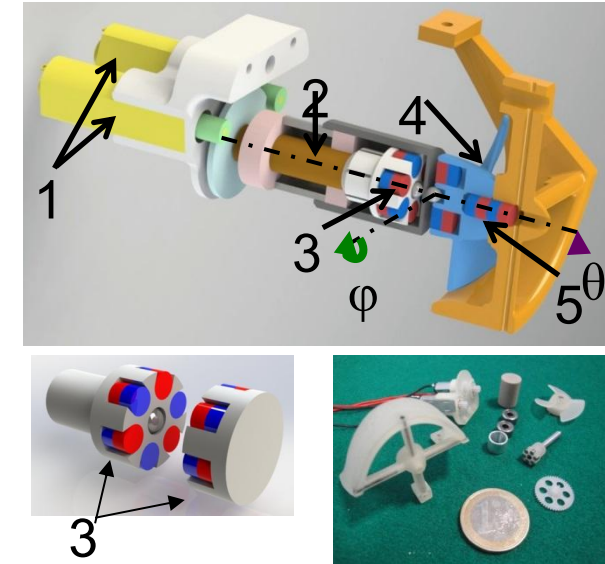
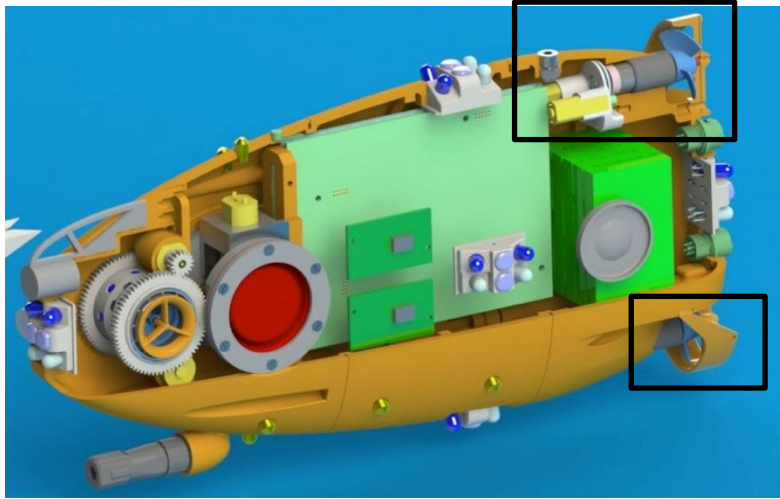
- 250 X 120 X 50 (mm)
- Surge 500 mm/s, Yaw 90° /s, Dive 80 mm/s up to 3 m
- 8 Li-Po cells with a capacity of 880mAh each.
- Blue lights and Electric fields communication.



- Pressure wave system
- Potential field system
- Blue Lights
- Pressure sensor



Propulsion System



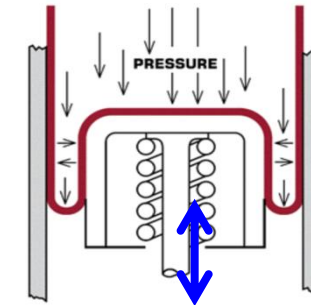
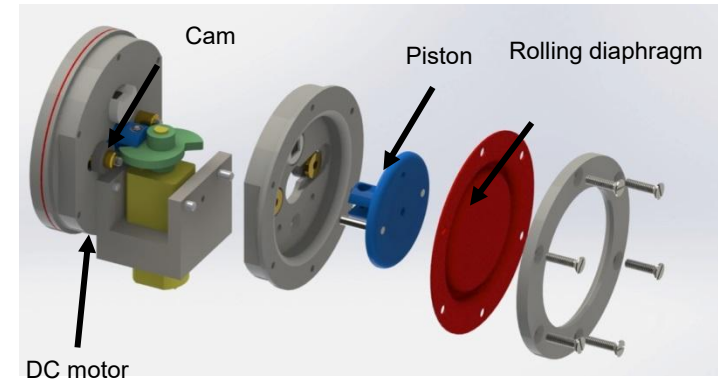
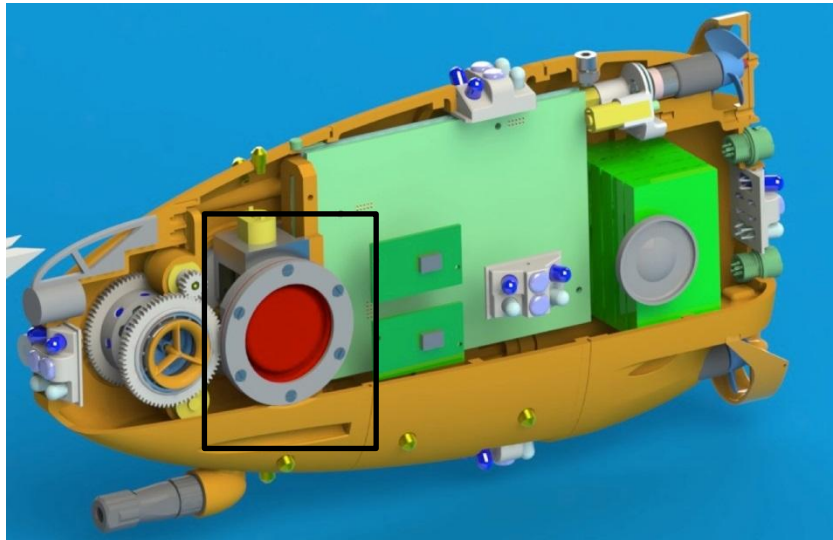
Trend of F_m and T_m as a function of the angle θ .

Main components

1. DC motors
2. Transmission shaft
3. Magnetic coupling
4. Propeller
5. Magnetic stabilization



Buoyancy system



Linear
displacement
($\pm 2 \text{ mm}$) of
the piston



Vertical velocity
0.08 m/s
Maximum depth:
3m

Advantages

- Reduced friction
- Waterproof



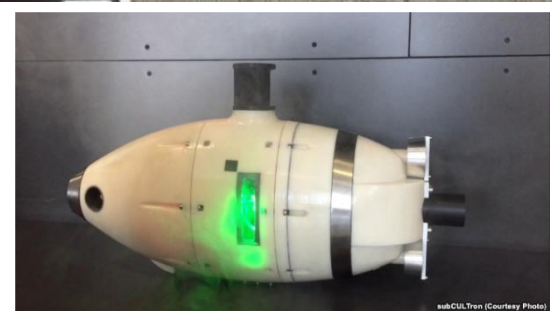


subCULTron

Submarine cultures perform long-term robotic exploration of
unconventional environmental niches



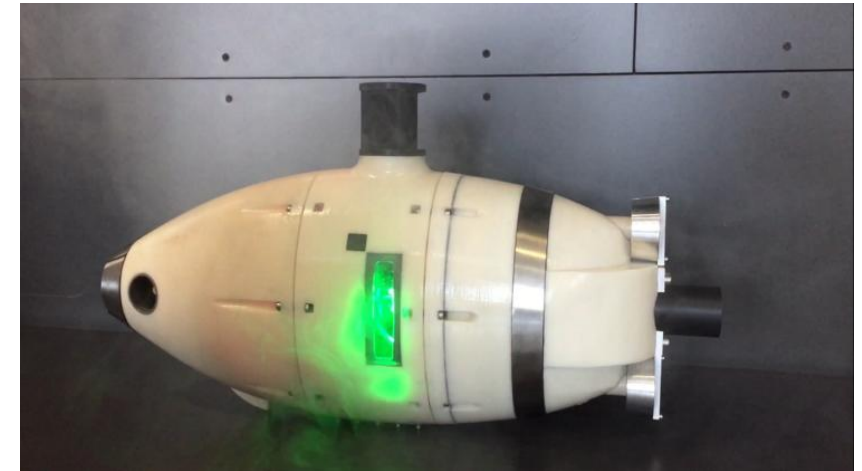
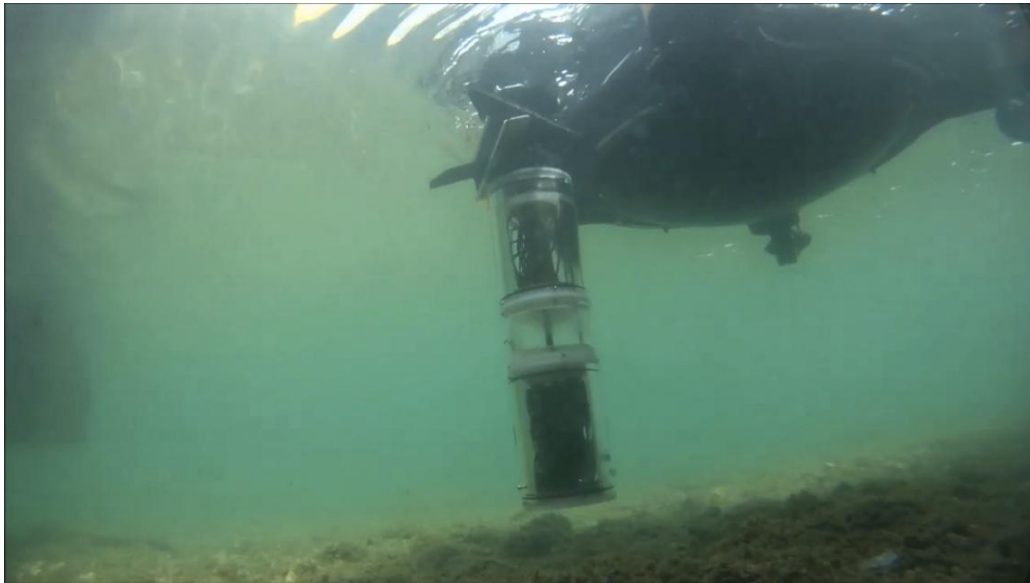
H2020-FETPROACT-2014



subCULTron

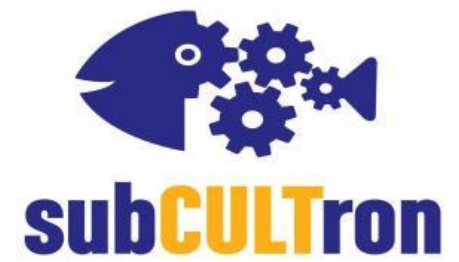
Submarine cultures perform long-term robotic exploration of unconventional environmental niches

subCULTron aims for achieving long-term autonomy in a learning, self-regulating, self-sustaining underwater society/culture of robots in a high-impact application area: Venice, Italy.



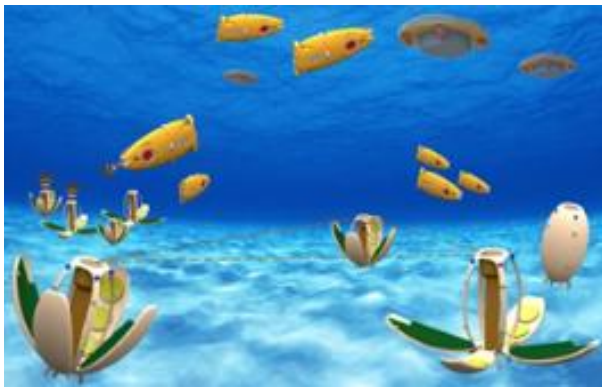


aMussel



Caratteristiche principali:

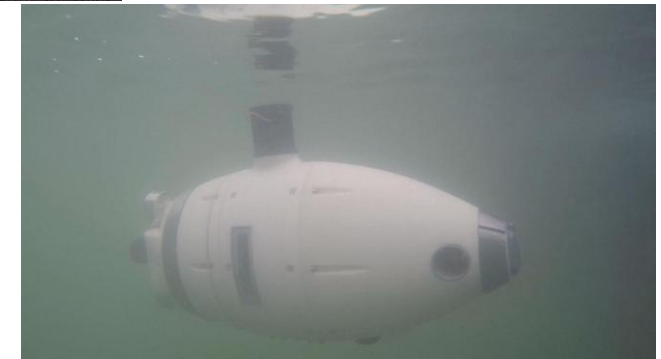
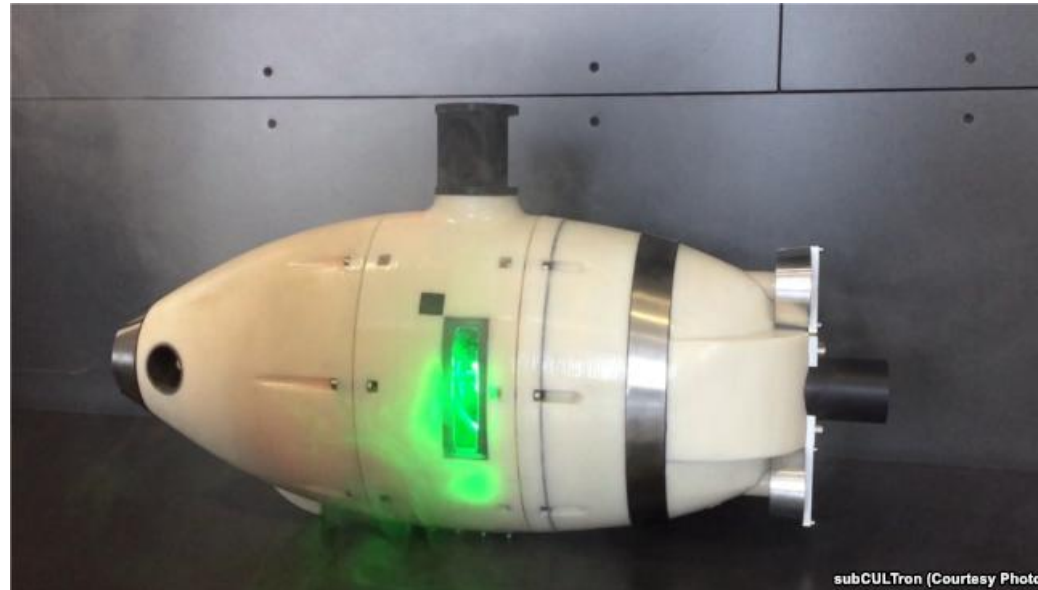
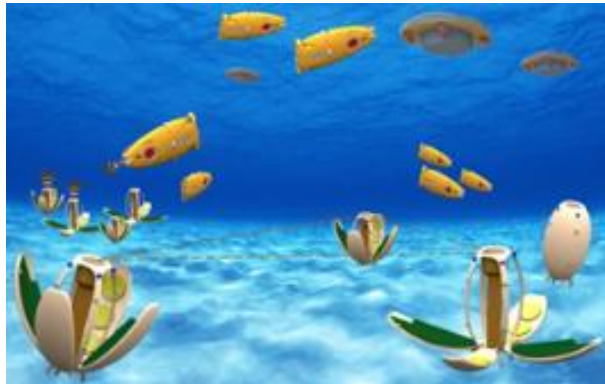
- Long term operation
- Diameter 120mm
- height 630mm
- Diving 15 m
- Energy autonomy 2 weeks
- Multiple communication channels



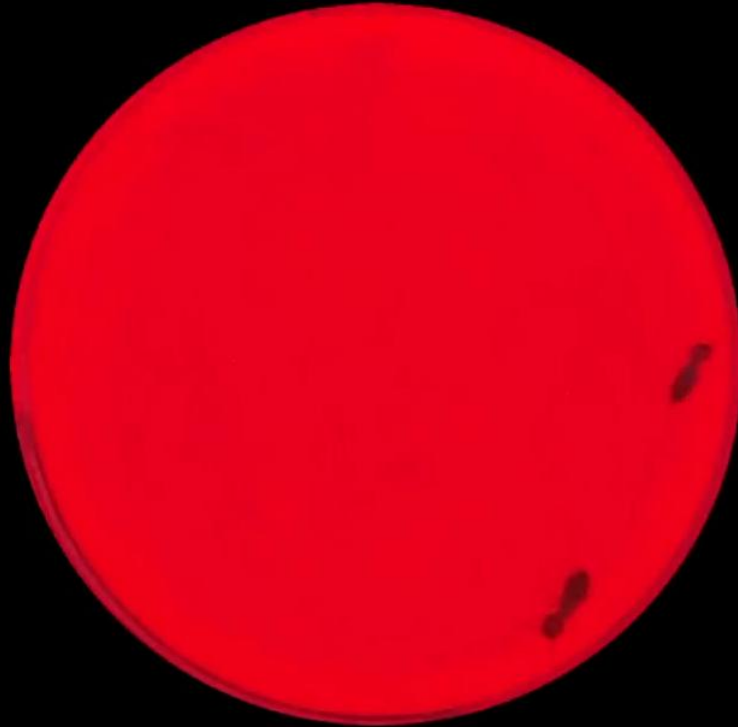
aFish

Caratteristiche principali:

- Long term operations
- Coordinate locomotion
- Swimming in the open sea
- Diving 15 m
- Energy autonomy 2
- Multiple communication channels



Super-organism
Self-organization
Information-flow
Collective behavior
Swarm intelligence
Stigmergy
Entropy



Transfer entropy
measures how much the
past of one process X
tells us about **the future**
of another process Y,
beyond what Y's own
past tells us.

$$T_{X \rightarrow Y} = I(X_{t-1}; Y_t | Y_{t-1})$$

Speed: 2.5x

Can we extract information from
collective behaviour in termites that
could be useful for swarm robotics?

worker  with  worker
soldier  with  worker
soldier  with  soldier